



Caddo iseries

Oscilloscopes. The current generation of oscilloscope is digital storage oscilloscope, digital phosphor oscilloscope, mixed-signal oscilloscope and mixed-domain oscilloscope. Analogue oscilloscopes too have become digital read-out type and offer various connectivity options.

Each type has its own distinct features. Digital storage oscilloscopes provide a high real-time sampling rate and memory depth. Besides, these perform a number of automatic measurements and display results on an LCD screen. With digital technique, they can be remotely operated and become a part of the automatic test setup.

Spectrum analysers. Spectrum analysers are available in frequencies ranging from 1 GHz to several GHz and are capable of a number of power calculations in different units with comparisons. EMI pre-compliance capability, limit lines, improvement in dynamic range and connectivity are some of the technical improvements.

Multimeters. Multimeters today are digital with LCD read-out. Human safety is of prime importance in both handheld and benchtop types. Resolutions up to 6.5, 7.5 and 8.5 digits are available.

New digital multimeter designs are increasingly expected to provide higher levels of precision and stability, as well as more standard built-in functions like capacitance, frequency and temperature measurement. End users are also looking for digital multimeters

that can ensure near-universal connectivity and come with simple software drivers, utilities and application-support software.

Digital multimeters are frequently being integrated with switch mainframes to provide more complete test and measurement solutions in a single enclosure.

Keithley's Series 3700A system switch/multimeter family is one example. "It offers scalable, instrument-grade switching and multi-channel measurement solutions that are optimised for automated testing of electronic products and components. Series 3700A mainframes are equipped with many standard features. For example, easy connectivity is supported with three remote interfaces: LXI/Ethernet, general-purpose interface bus (GPIB), and universal serial bus (USB). Fourteen digital I/O lines are also included, which are programmable," informs Chuck Cimino, marketing director, Keithley Instruments.

Arbitrary waveform generators. VLSI technology based arbitrary waveform generators offer a number of arbitrary waveforms in the built-in library.

Signal generators. Advancements in signal generator design include more programmable signal control, faster and deeper waveform memories, improved usability and connectivity, and greater throughput and signal performance. Other improvements include the simulation of high-speed serial interface clocks and data layers,

as well as LTE modulation evolution.

Power meters. With the growth of the smart grid, line power meters are increasingly required to provide serial communication capabilities, in addition to standard features like flexible connectivity, usability and application-support software. RF power meters must support higher frequencies and more selectivity due to increasingly complex communication modulation standards and tighter regulations on in-band and out-of-band transmitter power qualifications and testing.

Focus areas

The main technological advancements which are applicable to any T&M product are modularity or upgradeability, PC controllability and interconnectivity, application software, ruggedness, high bandwidths, accuracy, repeatability and multiple functionalities in the same instrument.

Also, the emphasis is on intelligent battery charging and power efficiency.

Above all comes the safety of operator and installation. "That's why the complete range of IDEAL's industrial clamps is designed with TightSight display. It allows an electrician to keep his head and face out of potentially dangerous panels while viewing the measurement. The high-voltage indicator is another feature that indicates presence of voltage even with the dial set to the wrong function. The clamps carry CAT III and IV ratings, and come in a rugged construction. UL listing adds to safety measures," says Sushil Tare, sales manager, electrical, Ideal Industries India.

There is a trend to use common development platforms on which a range of instruments can be built. This approach has several advantages for test system manufacturers and, in turn, their customers. It makes it more economical to deliver common interfaces and improved usability benefits. It also helps reduce development cycle times and provides greater economies of scale, which enable more product functionality and features to be offered at lower prices.

"Through shared components,